1. Can you explain the fundamental differences between supervised, unsupervised, and reinforcement learning? Provide a real-world example for each where you believe it's the most appropriate approach.
2. Describe a deep learning architecture you've worked with recently (e.g., Transformer, CNN, RNN, GAN). Explain its core components, its strengths, and a scenario where it excels.
3. Walk me through the typical steps involved in deploying a machine learning model into a production environment. What challenges have you faced in this process, and how did you overcome them?
4. When evaluating a classification model, beyond accuracy, what other metrics do you consider, and why? How would you choose between precision and recall for a specific business problem?
5. Data quality is paramount in AI. How do you approach data cleaning, feature engineering, and handling missing or imbalanced datasets in your projects?
6. How would you approach optimizing a computationally expensive AI model for faster inference or deployment on edge devices? What techniques come to mind?
7. Describe a time you encountered a significant bug or unexpected behavior in an AI model you developed. How did you diagnose the issue, and what was your resolution process?
8. What are some ethical considerations you keep in mind when developing AI systems, particularly concerning bias, privacy, or fairness? How do you try to mitigate these risks?
9. The field of AI is constantly evolving. How do you stay updated with the latest research, frameworks, and best practices?
10. Tell me about a project where you had to collaborate with a cross-functional team (e.g., data scientists, software engineers, product managers). What was your role, and how did you ensure effective communication and delivery?